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New Features of Web Clipping

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PalmSource
Conference 2000

Overview

- Major New Features
- Tags, Attributes, Special Strings
- INetLib
- UI
- Bug Fixes

Major New Features

- ‘C’ is for Cookies
- Color
- Open web browsing
- Localization
- Configurable Mailto:
- Lz77 Compression

Cookies

- Compliant with Netscape standard
  - Set-Cookie HTTP response header
    - Server sends Cookies for Cookie Jar
  - Cookie HTTP request header
    - Client sends Cookies from Cookie Jar
    - INetLIbSockHTTPReqSend () will query the jar





Cookies

- Cookie Jar on device
  - 50KB default upper size limit
    - 4KB limit to individual cookie size
  - Unused Cookies deleted
    - When cache is full
      - LRU Cookies deleted
      - A domain of cookies is deleted at this time
    - When expired
      - Auto-deleted when searching jar
      - Session-expiring Cookies deleted when exit WCA



Cookies

- INetLibSettingSet ()
  - Enable Cookies
    - InetSettingEnableCookies
  - Change size of Cookie Jar
    - InetSettingMaxCookieJarSize
    - Applies to all apps using INetLib





Cookies

- UI in Clipper preferences panel
  - Accept Cookies (checkbox)



Cookies sent

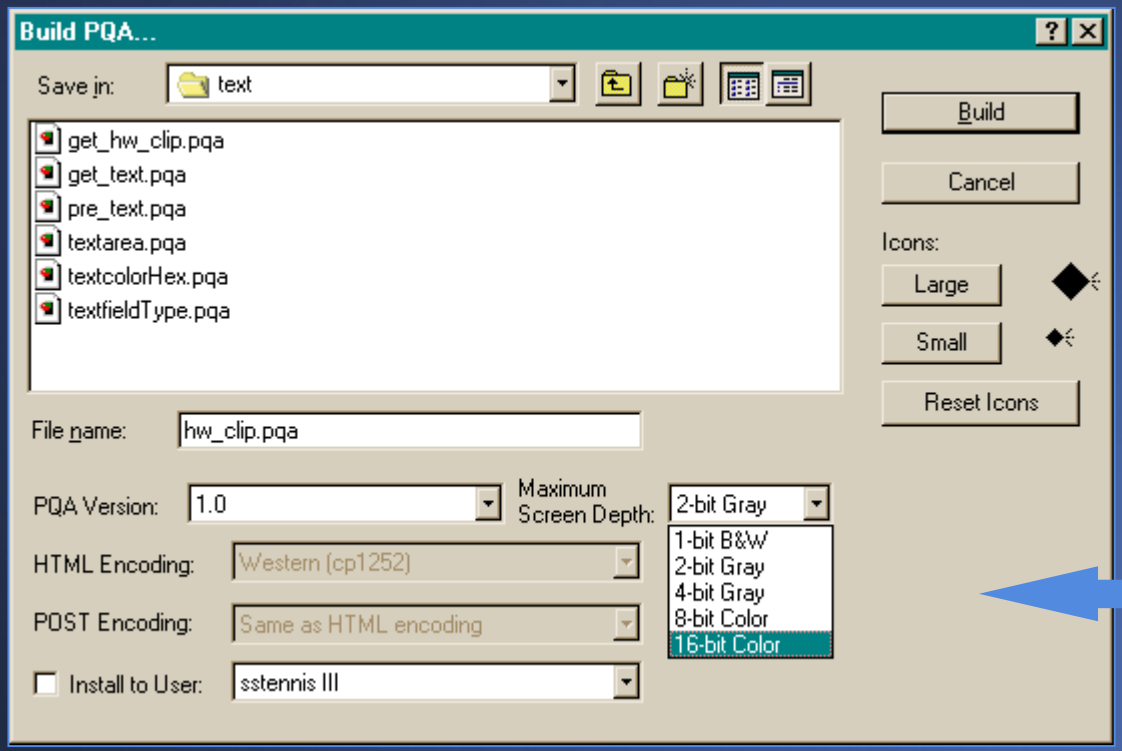
- Cookies not supported by OS < 4.0
  - WCP will not send cookies if device is not capable of handling them



- WCP server and Clipper support
  - Background color
  - Images
  - Text
  - 1, 2, 4, 8, 16-bit depth
  - Only the “Web safe” palette
    - Default Palm OS color palette
    - RGB values converted to closest match

Color

- WCA Builder
  - Image encoded for selected bit-depth



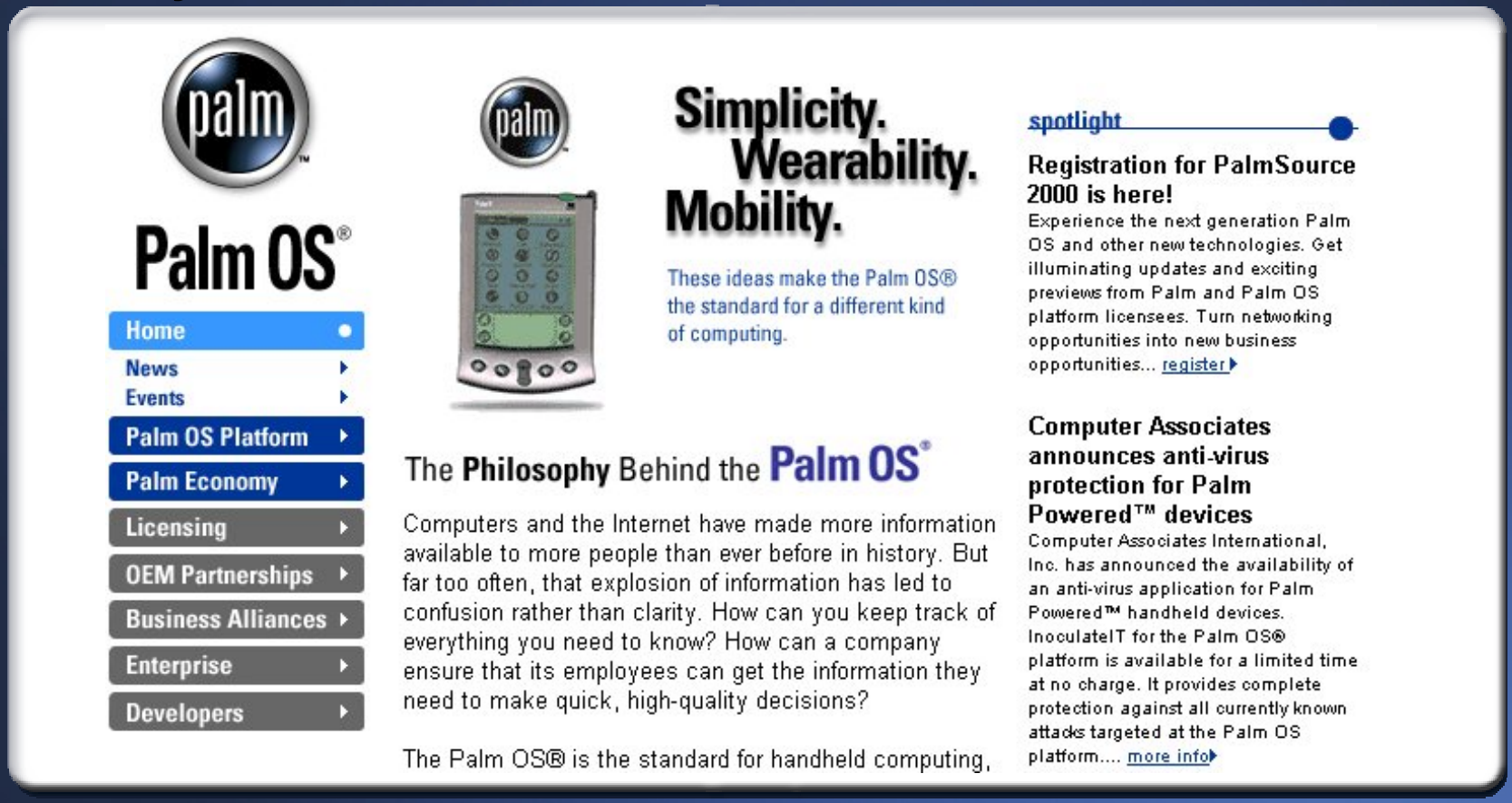
Choose Target Bit-Depth

Open Web Browsing

- Full Internet browsing
  - Scaling websites to make viewable on small screen
    - Images, Tables, Frames
    - Removing animation, video, audio
  - Makes the wireless experience more versatile and useable

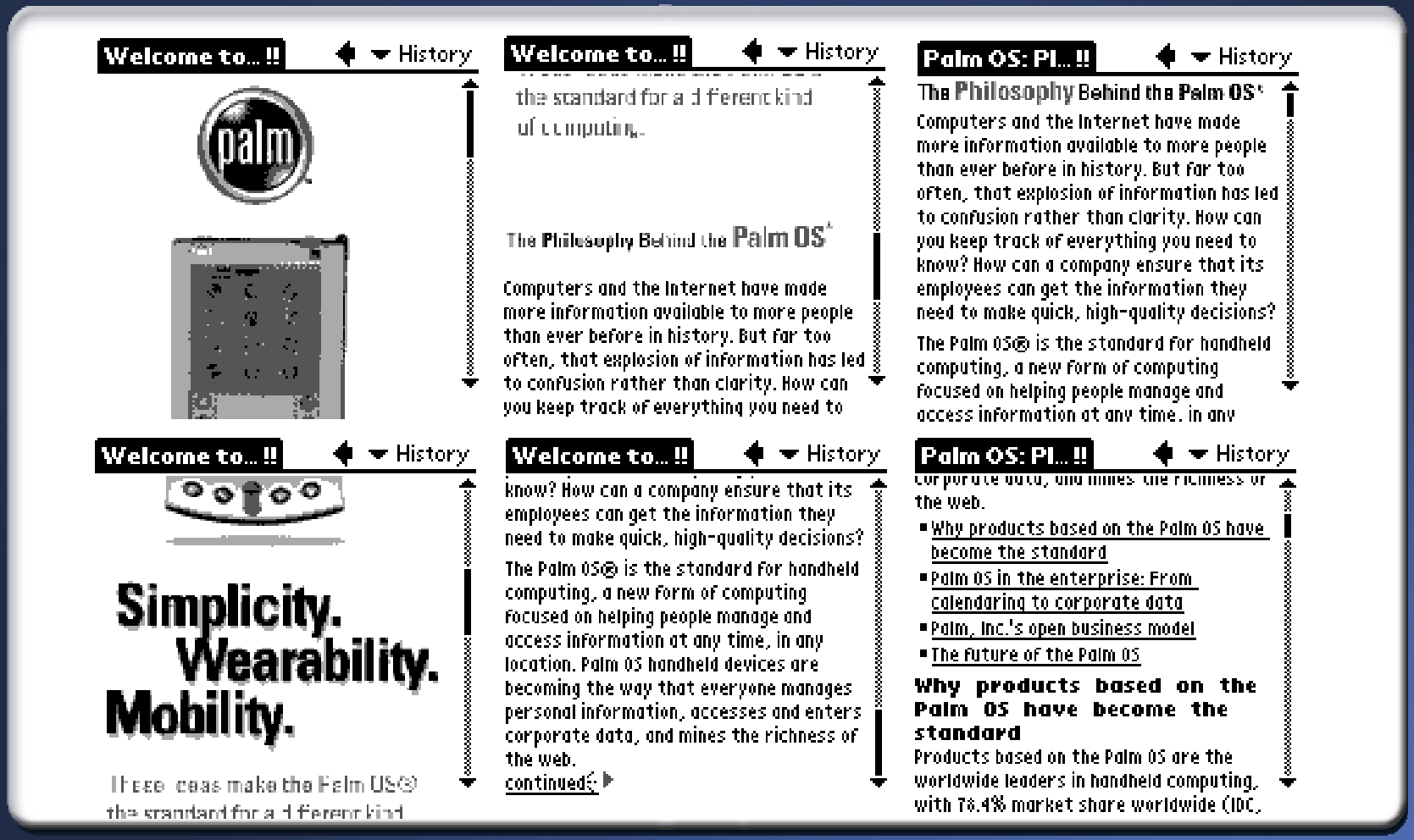
Open Web Browsing

- The reality is that a page that looks this way in a browser



Open Web Browsing

- Will be rendered as



Localization

- Clipper UI
- Error messages
  - WCP Server
  - Device
- 2-byte character support
  - Shift_Jis
    - Character set for Japanese localization
    - If device has supported character set, then content can be properly rendered
  - Device sends locale info to WCP server
    - Based on info in OS overlay

Configurable Mailto:

- Mailto: currently set to use iMessenger
- Now configurable
  - Register mail app with Exchange Manager

```
ExgRegisterData (CRID,          /* registering app */
                 exgRegSchemeID, /* URL scheme registry */
                 "mailto",       /* scheme to associate */
                 "Email URL",    /* description */
                  flags);
```

  - Exchange Mgr. launches registered default app
    - Default mail app is last mail app to have registered
  - UI for choosing mail app
    - Use “?mailto:”

Lz77 Compression

- Currently 5-bit compression used
  - Biased toward alphabets with lower case chars
- Lz77 better supports Shift_Jis
- No URL-encoding of data on upload/download
- Separate Library (not part of INetLib)

Lz77 Compression

- To use in a PRC
  - *ctpConvNoneLZ77* INetLib conversion algorithm
    - INetLibSettingSet ()
    - Download data only
    - Decompress download data using Lz77 API
  - Sample code in the knowledge base

Tags, Attributes, Special Strings

- PalmHREFStyle and PalmStyle
- PalmDoNotCache
- PalmBinary
- %Location:…
- %WCDevCaps

Tag—PalmHREFStyle Attribute—PalmStyle

- Full URLs downloaded
  - Avoid errors related to link hashing
  - 2 values
    - Full
    - Indexed (default)

- All links in clipping

```
<meta name="PalmHREFStyle" content="full">
```

- Individual link

```
<a href="http://www.palmos.com/" PalmStyle="full">
Full Link</a>
```

Tag—PalmDoNotCache

- Prevent storage of clipping in Clipper cache
  - Prevent device-side caching of sensitive data
  - Won't show-up in History List

```
<meta name="PalmDoNotCache" content="true">
```

- Disables type ahead for that clipping
  - Prevents storage of entered credit card data, etc.

Attribute—PalmBinary

- Download a binary file
  - 5 KB safe limit for Mobitex networks

Get App

- File handled by Exchange Mgr.
  - File sent to app registered to handle file type
    - Based on MIME type, extension, etc.

Special String—%Location

- Carrier provides data
  - Raw network data taken at time of request
  - Carrier provides service provider with map table
- WCP server maps raw data to location info
  - Location info desired indicated by code(s)
    - Codes in any order, almost any separator
    - End of list indicated by '.'
  - String not converted if no location data available
  - Location info URL encoded (to content server) e.g.:
    - Given ?loc=%location:z;c.
    - Sent as ?loc=95002%3bSanta+Clara

Special String—%Location

- Location info codes

Type of Location Information	Location Code
Zip code and Postal code	z
City name	c
Country code	Y
Country name	y
State name (large political division)	l
State code (large political division)	L
County name (small political division)	o
County code (small political division)	O
Global Positioning System (lat/long)	J
Raw location data (hexadecimal)	R

- Country code (**Y**)
  - Based on *Codes for the Representation of Names of Countries (ISO 3166-1993 (E))*
    - American National Standards Institute, Inc.

Special String—%WCDevCaps

- Why?
  - Many different Palm OS devices accessing Internet
  - Some new features not backward compatible
- Sends data on OS, screen depth, etc.
- Used like %Location and %Zipcode
- Replaced by WCP server
  - HEX value, low bytes first
  - Not replaced if not supported

INetLib

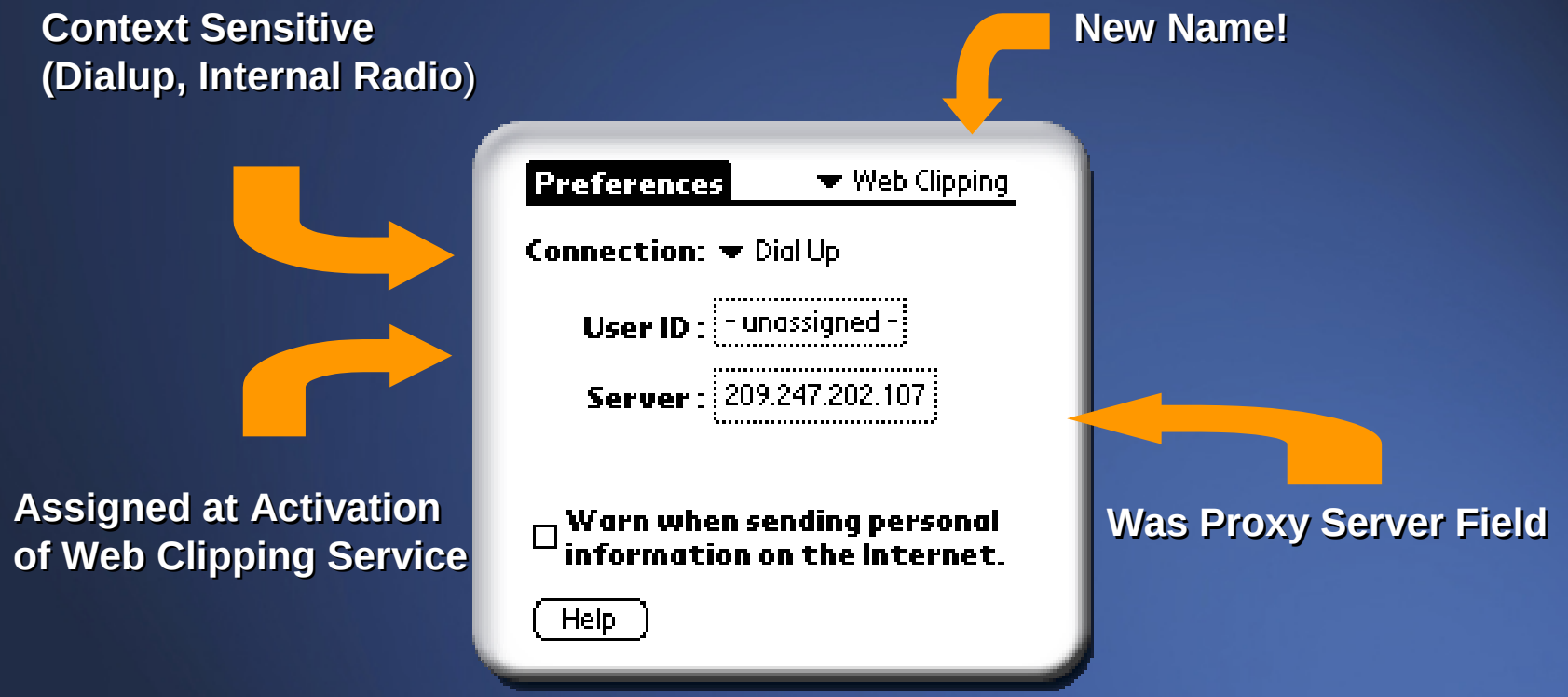
- Content-type HTTP request setting
  - Upload POST data without URL encoding
  - Remember:
    - Content string ptr must not be NULL
    - "application/octet-stream" used for empty content string
    - Beware sending type not recognized by destination server
- Previously mentioned
  - Cookies and Lz77

UI

- Web Clipping Preferences Panel
- Menu Features
- Command Bar
- Type Ahead
- Horizontal Scrollbar
- Cue for Encrypting
- Secure to Non-Secure Warning

Web Clipping Preferences Panel

- Was Wireless Prefs panel



Menu Features

- Context Sensitive Menus
- New Menus
  - Bigger/Smaller Font
    - Scale all text up or down
    - State retained while Clipper active
  - Disconnect
    - PPP connections only
  - Clipper Prefs Panel
    - Next slide



Menu Features

- Clipper Prefs Panel



None / Quality vs. Speed

Send to/from Device

Set to 0 to
Clear Cache

Default:
(2 or 8) MB RAM x 25KB

e.g.: 2 x 25KB = 50KB

Command Bar

- Context sensitive
  - Cut, Copy, Paste, Font change, Disconnect



    - Used with forms with editable fields
    - Disconnect is present only if the IP connection is open
  - Font change, Disconnect



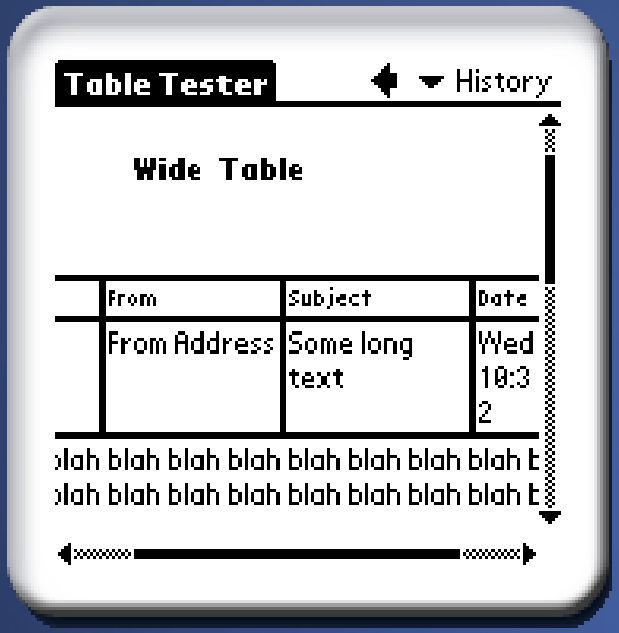
    - Used in forms without any editable fields
    - Copy Page is only option for capturing data

Type Ahead

- Stores entered values
  - Only when form is submitted
  - DOES NOT store password field values
  - 1 db for all entries in WCAs
    - Values sorted by most recently used
    - 100 values max
  - Disabled by
    - UI checkbox (Easter Egg only)
    - PalmDoNotCache meta tag
- Match stored values against what user types
  - 1 character type ahead matching

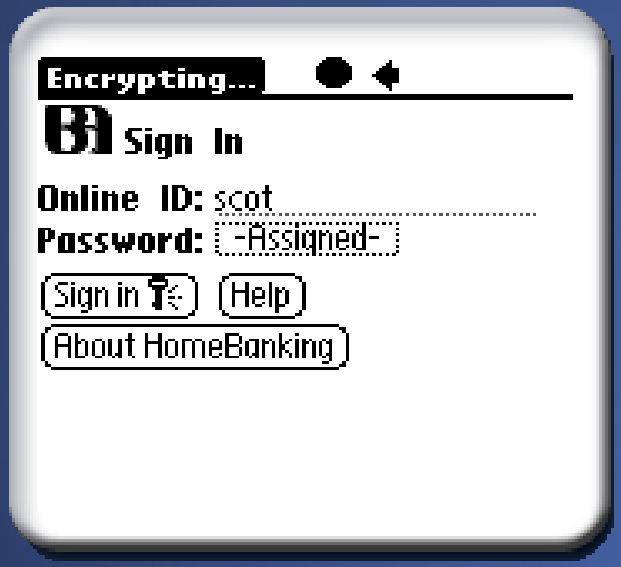
Horizontal ScrollBar

- Wide content sometimes desirable
  - e.g.: Data best shown with a wide table
- Automatically enabled for Palm Friendly pages



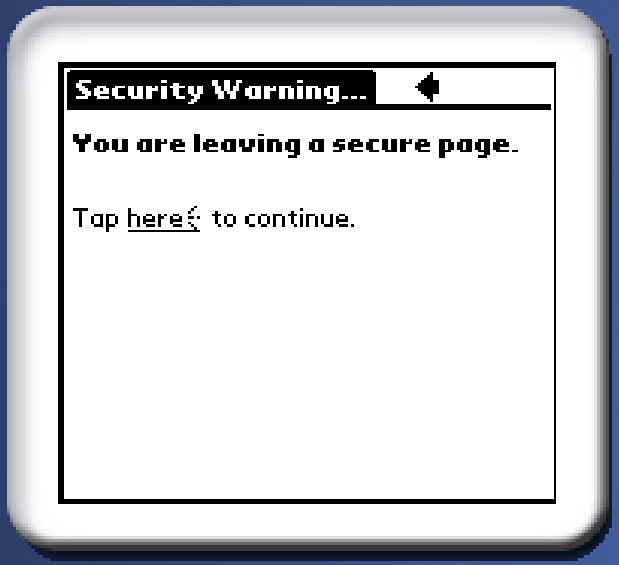
Cue for Encrypting

- No cue currently
  - Noticeable pause while encrypting
- Assure user of security with a simple cue



Secure to Non-Secure Warning

- Redirected from a secure to a non-secure URL
- Warning sent to device
  - Enables user to make informed choices
  - *Here* link is to the URL requested



Bug Fixes

- Proxy firewall and emulator problem
- Guarantee support for rendering at least
  - 16 KB for 2MB RAM device
  - 32 KB for 8MB RAM device
- Self-referencing links
- '=' now follows all empty form field names
- Others


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